REVIEW FOR FINAL EXAM GRADE 8

A. Multiple choice questions

1. Which of the following is the formula of acceleration?

a. $a = \frac{d}{t}$ **b.** $a = v \times d$

c. $a = \frac{v_2^2 - v_1^2}{t_2 - t_1}$ **d.** $a = \frac{v_2 - v_1}{t_2 - t_1}$

2. Which of the following is the velocity of an object?

a. $v = at$ **b.** $v = d/t$

c. $v = d \times t$ **d.** $v = \frac{v_1 + v_2}{2}$

3. Which of the following is the rate of change of velocity with time?

a. Acceleration **b.** Speed
c. Displacement **d.** Vector

4. You stretch a spring by pulling one end of it to the right. Which best describes the force exerted on the spring?

a. Compression force to the left.
b. Compression force to the right.
c. Tension force to the left.
d. Tension force to the right.

5. Which of the following is the SI unit of weight?

a. Newton per meters (N/m) **b.** Kilogram (kg)
c. Newton (N) **d.** meter per second squares (m/s^2)

6. The Newton's first law states that:

a. When the net force acting on an object is nonzero, if the body is at rest, it will be at rest. If it is moving, it continues with a constant velocity.
b. When the net force acting on an object equal zero, if the body is at rest, it will be at rest. If it is moving, it continues with a constant velocity.
c. When the net force acting on an object equal zero, if the body is at rest, it will start moving, if it is moving, it continues with a constant acceleration.

7. A book is on a table. How many forces exert on the book? What are they?

a. Two forces, gravitational force and normal force.
b. ONLY 1 force, gravitational force
c. THREE forces, gravitational force, a push, and normal force.

8.The brakes are applied to a train travelling at 25 m/s and it takes 18 s to stop with uniform deceleration. What is its deceleration?

a. $a = 1.39\,m/s^2$ b. $a = -1.39\,m/s^2$
c. $a = 450\,m/s^2$ d. $a = -450\,m/s^2$

10. What is the area under the velocity-time graph represents?

a. Acceleration **b.** Distance
c. Velocity **d.** None of these

11. What is the slope of a line on a distance time graph?

a. displacement **b.** speed
c. force **d.** acceleration

12. In velocity-time graph, distance travelled is represented by

a. the slope of the graph. **b.** the gradient of the graph
c. the area under the graph. **d.** None of the above.

13. Which of the following is true about an object in free fall?

a. Its acceleration depends on its mass.
b. It has no inertia.
c. It pulls on Earth, and Earth pulls on it.
d. Its momentum is constant.

14. How can speed be defined?

a. acceleration/time **b.** change in velocity/time
c. distance/time **d.** displacement/time

15. Sound travels at a speed of 330 m/s. How long does it take for the sound of thunder to travel 1,485 m?

a. 45 s **b.** 4.5 s
c. 4,900 s **d.** 0.22 s

16. The graph shows how a cyclist's speed changed over 0.5 h. What is the cyclist's average speed if the trip was 6 km?

a. 3 km/h **b.** 10 km/h
c. 12 km/h **d.** 20 km/h

17. A skier is going down a hill at a speed of 9 m/s. The hill gets steeper and her speed increases to 18 m/s in 3 s. What is her acceleration?

a. 9 m/s^2 **b.** 3 m/s^2
c. 27 m/s^2 **d.** 6 m/s^2

18. A car starts from rest and uniformly accelerate s to a final speed of 20.0 m/s in a time of 15.0 s. How far does the car travel during this time?

a. 150 m **b.** 300 m
c. 450 m **d.** 600 m

19. A car moves 0.80 m in 1.0 s at the constant velocity. If it continues, how far will it travel in 3.0 s?

a. 2.4 m **b.** 3.6 m
c. 4.8 m **d.** 7.2 m

20. Which of the following is NOT a force?

a. weight **b.** momentum
c. friction **d.** air resistance

21. The gravitational acceleration the acceleration of bodies when they are in

a. ground **b.** air
c. free falling **d.** all of these

22. Which of the following is the definition of force?

a. is the push or pull on an object.
b. is the matter in an object.
c. is the mass of an object.
d. is the moving of an object

23. The velocity of ball at highest point when it thrown vertically upward is

a. 0 m/s **b.** 10 m/s
c. 1 m/s **d.** None of these

24. The height of tower when stone dropped from it reaches ground in 5 s is

a. 130 m **b.** 125 m
c. 150 m **d.** 200 m

25. Projectile will attain its maximum range, if it is fired at an angle of

a. 30^0 **b.** 47^0
c. 90^0 **d.** 45^0

26. As the balls towards the ground, its velocity

a. increases **b.** decrease
c. remains constant **d.** becomes zero

27. If initial velocity of an object is zero, then distance covered by it in time and acceleration of 9.8 m/s^2 would be

a. $h = 2.9t^2$ **b.** $h = 3t^2$
c. $h = 4t^2$ **d.** $h = 4.9t^2$

28. A body of mass 1 kg has weight 10 N at a certain place. What is the weight of a 100kg a) 100 kg, b) 5kg, c) 50 kg?

a. $a).W = 100N, b).W = 50N, c).W = 5N$
b. $a).W = 10N, b).W = 50N, c).W = 500N$
c. $a).W = 100N, b).W = 50N, c).W = 500N$
d. $a).W = 1000N, b).W = 50N, c).W = 500N$

29. What is the force constant of a spring which is stretched a). 4 mm by a force of 6.0 N, b). 8 cm by a mass of 400g?

a. $a).k = 1500\,N/m, b).k = 50\,N/m$
b. $a).k = 1500\,N/m, b).k = 666.67\,N/m$
c. $a).k = 150\,N/m, b).k = 50\,N/m$
d. $a).k = 1.5\,N/m, b).k = 500\,N/m$

30. Two forces on an object are in the same direction therefore, the resultant force is

a. takes away one from other **b.** multiply them
c. divide one by the other **d.** add them together

31. Which of the following is obeyed the first law of Newton?

a. v = constant, a =0 **b.** a = constant, v = 0
c. v = constant, a = constant **d.** None of these

32. Which of the following is the definition of inertia?

a. is the push or pull on an object.
b. is the quantity specify how strong of an object.
c. is the measure of inertia
d. is a tendency of an object to resist the change in its motion.

33. Calculate gravity on Mars Find the acceleration due to gravity on Mars if a person with a mass of 60.0 kg weighs 222 N on Mars?

a. $g = 1.66\,N/kg$ **b.** $g = 9.8\,N/kg$
c. $g = 3.7\,N/kg$ **d.** None of these

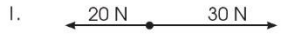
34. You push a friend on a sled. Your friend and the sled together have a mass of 70 kg. If the net force on the sled is 35 N, what is the sled's acceleration?

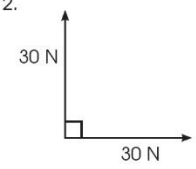
a. $a = 0.5\,m/s^2$ **b.** $a = 2\,m/s^2$
c. $a = 0.2\,m/s^2$ **d.** $a = 5.0\,m/s^2$

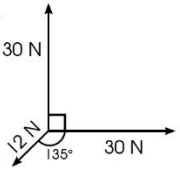
35. If a force of 50 Newton's was applied to an object with a mass of 5 kilograms, what will the object's acceleration be?

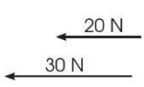
a. $a = 10\,m/s^2$ **b.** $a = 1\,m/s^2$
c. $a = 100\,m/s^2$ **d.** None of these

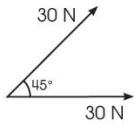
35. Find the resultant force in each diagram and draw the resultant vector. Use a ruler and a protractor where necessary. Scale: 1 cm = 10 N, where N represents newtons of force.

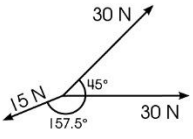
1. 

2. 

3. 

4. 

5. 

6. 

B. Matching question

36. Match the word with the correct definition or corresponding phrase.

1.___ Acceleration	A. is the force of gravity on the body.
2. ___ Velocity	B. is the length of path.
3. ___ Distance	C. is the push or pull on the object.
4. ___ Gravity	D. is the SI unit of force.
5. ___ Force	E. is rate at which an object changes its velocity.
6. ___ The Newton	F. is the total distance travelled per unit time.
7. ___ Weight	G. is the combining or adding between two or more than two forces into a single force.
8. ___ Hook's law	F. is the tendency of an object to resist the change in its motion.
9. ___ Resultant force	H. is the force oppose one surface moving, or trying to move over another.
10. __ Projectile motion	I. the extension is proportional to stretching force.
11. __ Mass	J. when the object is given an initial velocity.
12. __ Inertia	J. is the measured of inertia.
13. __ Friction force	K. is the acceleration of free fall which has the same direction for objects falling.

C. Work problems

37. Find the distance traveled by a car that travels with an average speed of 110 km/h for 3.5 h.

38. An elevator travels a distance of 220 m from the first floor to the 60th floor in 27.5 s. What is the elevator's average speed?

39. A motorcyclist travels with an average speed of 20 km/h. If the cyclist is going to a friend's house 5 km away, how long does it take the cyclist to make the trip?

40. A police car accelerates from 15 m/s to 35 m/s in 5 seconds. Calculate the acceleration.

41. A rocket has a mass of 600kg. At lift-off the rocket engine exerts an upward force of 30000N. What is the resultant force on the rocket?

42. A spring is stretched 0.02m by a weight of 4.0N. Calculate the weight of a new object that causes an extension of 4cm.

43. A car accelerates from a standstill to 60 km/h in 10.0 seconds. What is its acceleration?

44. A car accelerates from 25 km/h to 55 km/h in 30 seconds. What is its acceleration?

45. A train is accelerating at a rate of 2.0 km/h/s. Its initial velocity is 20 km/h.
What is its velocity after 30 seconds?

46. A runner achieves a velocity of 11. l m/s 9 seconds after he begins. What is his acceleration?

47. What is the velocity of a quarter dropped from a tower after 10 s?

48. If a block of wood dropped from a tall building has attained a velocity of 78.4 m/s, how long has it been falling?

49. If a ball that is freely falling has attained a velocity of 19.6 m/s after 2 s, what is its velocity 5 s later?

50. A piece of metal has attained a velocity of 107.8 m/s after falling for 10 s. What is its initial velocity?

51. How long will it take an object that falls from rest to attain a velocity of 147 m/s?

52. How far will a rubber ball fall in 10 s?

53. How far will a rubber ball fall in 20 s?

54. How long will it take an object dropped from a window to fall a distance of 78.4 m?

55. What is the final velocity of the ball in problem 52?

56. What is the average velocity of the ball in problem 52?

57. An airplane is traveling at an altitude of 31,360 m. A box of supplies is dropped from its cargo hold. How long will it take to reach the ground?

58. At what velocity will the box in problem 57 be traveling when it hits the ground?

59. Name the energy transfers which occur when
a) an electric bell rings,
b) someone speaks into a microphone,
c) a ball is thrown upwards,
d) there is a picture on a television screen,
e) a torch is on.

60. A hiker climbs a hill 400 m high. If she has a mass of 60 kg calculate the work, she does in lifting her body to the top of the hill.

61. In loading a lorry a man lifts boxes each of weight 200 N through a height of 3.5 m.
a) How much work does he do in lifting one box?
b) How much energy is transferred when one box is lifted?
c) If he lifts four boxes per minute at what power is he working?